Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc. I.T Semester V)

Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T018	Name: Sachin Ramesh Mehta
Class: TYIT	Batch: 1
Date of Assignment: 04/08/2026	Date/Time of Submission: 04/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code:

Offline.html:

```
<!DOCTYPE html>
<html>
<head>
   <title>Offline</title>
</head>
<body>
<header>
   <h2>My College Website</h2>
</header>
<main>
   <h2> You are Offline</h2>
   <p>Please check your internet connection.</p>
   <button onclick="location.reload()">Try   Again</button>
</main>
<footer>
   <p>© 2026 My College Website</p>
</footer>
</body>
</html>
```

Service-worker.json:

```
const CACHE = "v1";
self.addEventListener("install", e => {
   e.waitUntil(
      caches.open(CACHE).then(cache => {
         return cache.addAll([
            "/",
            "/offline.html"
         ]);
      })   );



   console.log("Service Worker Installed");
```

```
});
self.addEventListener("fetch", e => {
    e.respondWith(
        fetch(e.request).catch(() => caches.match("/offline.html"))
    );
});
```

_Layout.cshtml:
Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

Before </body> tag:

```
<script>
if ('serviceWorker' in navigator) {
navigator.serviceWorker.register('/service-worker.js');
}
</script>
```

Output:

b. Create a Student Record Management PWA.

Code:

Index.cshtml:

```
    <h1>Offline Page</h1>
    <h2>Student Record</h2>

    <p>Roll No : @Model.RollNo</p>

    <p>Name : @Model.Name</p>
```

Student.cs:

```
public class student
{
    1 reference
    public int RollNo { get; set; }
    1 reference
    public string Name { get; set; }
}
```

HomeController.cs:

```
student s = new student();

s.RollNo = 101;
s.Name = "Sachin";

return View(s);
```

service-worker:

```
const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});

self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});
```

Output:

Online page:

Welcome

Learn about building Web apps with ASP.NET Core.

Offline Page

Student Record

Roll No : 101

Name : Sachin

Offline page:

WebApplication3

- Home
- Privacy

Welcome

Learn about building Web apps with ASP.NET Core.

Offline Page

Student Record

Roll No : 101

Name : Sachin

© 2026 - WebApplication3 - Privacy